GRADE 3 · DEBUGGING

Extra or Missing Block

Ontario Math C3.2 — read, find, and fix code

Name: .... Date: ....

★ **I can fix a loop that has an extra block or a missing block.**

Too many or too few blocks

Some bugs are not about numbers — they are about the BLOCKS inside the loop. There might be an EXTRA block, or a needed block might be MISSING.

Read what Bit should do, then check the inside blocks: is one extra, or is one missing?

Bit should do clap, stomp — but this has an EXTRA block

```
repeat 3
    clap
    stomp
    stomp
```

There is an EXTRA stomp. Bit does 6 stomps instead of 3. Remove the extra block.

1 Find the extra block

The loop should be clap, stomp — but it has clap, stomp, stomp. Which block is extra? How many stomps does Bit do now instead of 3? ______

____

____

2 Find the missing block

This should draw a SQUARE: for i in range(4): bit.forward(50). But Bit draws a straight line! What block is MISSING inside the loop?

____

____

3 Write the fix

Add the missing block to fix the square. Write the line that goes after bit.forward(50):

bit.________(______)

___

4 Challenge

A square loop has an EXTRA turn: for i in range(4): bit.forward(50); bit.right(90); bit.right(90). Bit just goes back and forth on a line. Why? What should you remove?

___

___

Big idea

Count the blocks inside the loop, not just the repeats. An extra block does too much; a missing block leaves the job unfinished.

TEACHER GUIDE · GRADE 3 · DEBUGGING

Extra or Missing Block

Quick facilitation notes & answer key

What this worksheet teaches

Students debug loops with an EXTRA inside block (does too much) or a MISSING inside block (job unfinished), in block and shape cases, and describe how adding or removing one block fixes the result. The method here is SHOULD → DOES → FIX: say what the code SHOULD do, run it to see what it DOES, then FIX the one thing that is wrong. No coding experience needed.

Before you start

No computer needed — the worksheet shows the code and what Bit should do. Optional: a Scratch-style app (for the block loops) or a free Python-turtle playground (for the shapes) to run the code — you only press Run.

How to lead it (about 30 minutes)

1. Show it (you do). An extra block makes Bit do too much; a missing block leaves the job unfinished.
2. Do one together. Exercise 1 — the second stomp is extra.
3. Let them work. Students add the missing turn to close a square (Ex 2–3) and remove an extra turn (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — the second stomp is extra; Bit does 6 stomps (remove one to get 3). Exercise 2 — the turn block (turn right 90) is missing.

Exercise 3 — turn right 90. Exercise 4 (challenge) — two turns of 90 add up to a half-turn each side, so Bit goes back and forth; remove the extra turn right 90.

If students get stuck — and ways to adjust

Watch for: changing a block instead of adding or removing one. Decide first: too much, or unfinished?

Make it easier: count the blocks the goal needs versus the blocks inside.

Make it harder: ask what an extra forward (instead of an extra turn) would do to a square.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, find what is wrong, fix it, and explain what the change did.

You'll know it worked when

The child adds the missing block or removes the extra one to match the goal.